Fall 2025 — Course Project Report

VISUAL ANALYTICS

The Global AI Landscape

2010-2024

RONG ZENG · STUDENT ID: 2244780 · ERASMUS

Sapienza University of Rome

CONTENT

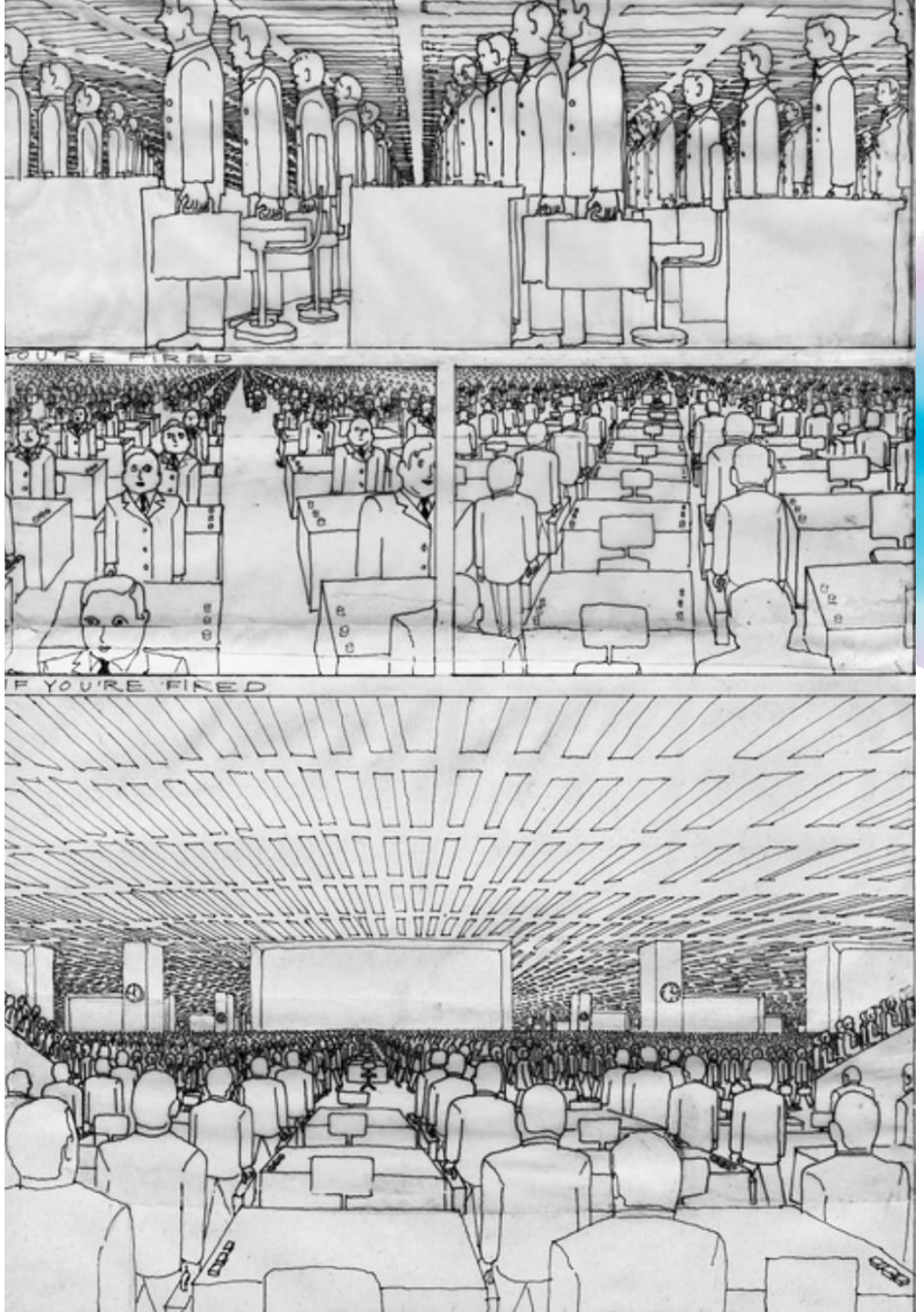
Part.01
Introduction
Overview of global AI capability differences

Part.02
Datasets and domains
Integrated country–year data from multiple sources

Part.03
Solution
Interactive dashboard with PCA, clustering, and linked views

Part.04
Insight
Key patterns in AI capability, compute, and policy

Part.05
Limitations
in data quality, coverage, and imputation



Introdation

This project explores global AI capability differences across countries by combining research, compute, policy, and development indicators from 2010 to 2024

Artificial intelligence is growing quickly, but countries do not develop at the same speed.

To better understand these differences, this project builds a Visual Analytics system for 30 countries from 2010 to 2024. The system combines data on AI research, compute capacity, energy, and policy into an interactive dashboard.

The main users of this tool are policy analysts, researchers, students, and job seekers interested in global AI development. The system helps them compare countries, explore AI capability gaps, and understand global trends in research, compute, and policy.

Part 1

Part 2

Variable Groups and Data Sources

AI Capability & Innovation: AI publications, R&D, researchers
Source: OECD.AI; World Bank

AI Compute & Infrastructure: TOP500 systems, Rmax PFLOPS, electricity use
Source: TOP500; World Bank

AI Governance & Policy: AI policy initiatives
Source: World Bank

ICT Infrastructure: internet use, broadband, mobile subscriptions
Source: World Bank

Socioeconomic Context: GDP per capita, income group, OECD/GPAI membership
Source: World Bank; OECD; GPAI

Part 2

Date Preparation

MY WORK FLOW

Step 1 — Source Cleaning

World Bank indicators
TOP500 data
AI publications
AI policy records

Step 2 — Harmonization

Standardized country names
Aligned years: 2010–2024
Merged into one master table

Step 3 —Missing Value Treatment

Kept meaningful zeros unchanged
Interpolation + ffill + bfill for continuous variables
KNN imputation for ai_publications

Step 4 —Analysis-Ready Outputs

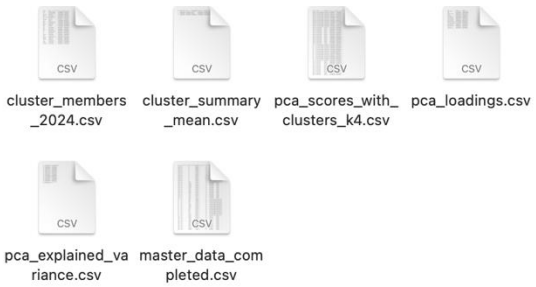
z-score normalization before PCA
PCA + KMeans outputs
Final frontend tables for dashboard

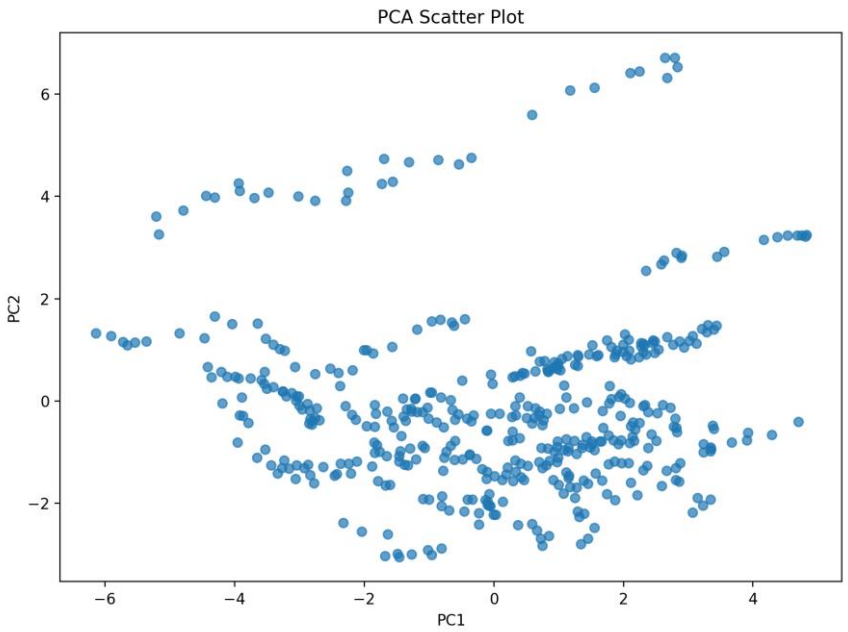
Step 3 - Missing Value Treatment

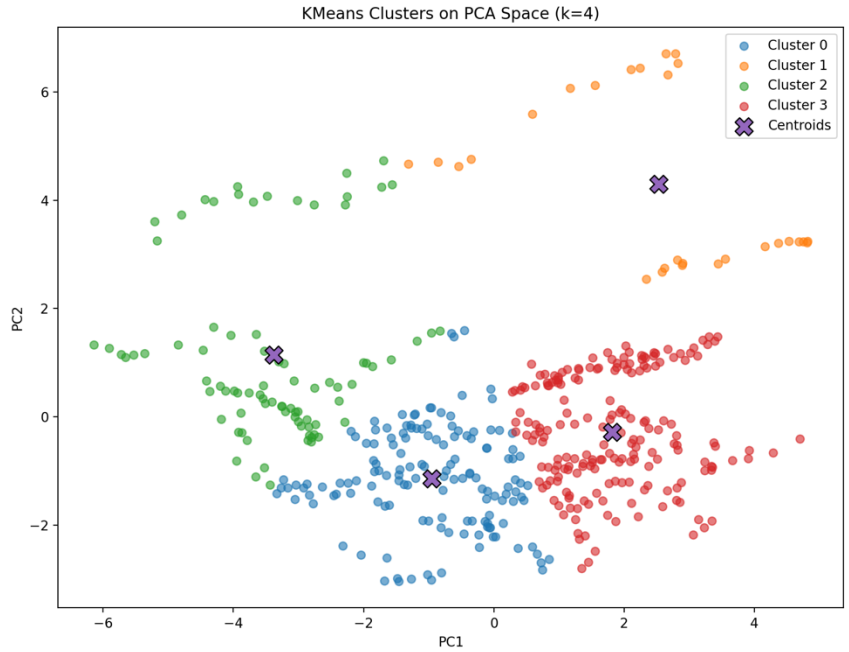
Variable Type	Method
Identifiers / meaningful zeros	Kept unchanged
Continuous country–year indicators	Linear interpolation + ffill + bfill
ai_publications	KNN imputation

Step 4 — Analysis-Ready Outputs

- Standardized numeric variables with z-score normalization before PCA
- Generated PCA scores and KMeans cluster labels
- Exported final analysis files for visualization and interaction







Technologies Used

- HTML for page structure
- CSS for layout and styling
- JavaScript for interaction logic
- D3.js for interactive visualizations
- Python for data preprocessing, PCA, and KMeans
- Local Python HTTP server for running the dashboard

Part 3

My SOLUTION .

Dashboard Panels

PCA Scatter Plot
main view for country comparison

Ranking Bar Chart
coordinated ranking view

Parallel Coordinates
multivariate cluster profile view

Trajectory View
temporal evolution of selected countries

Summary Statistics
interaction-triggered analytics panel

Dashboard Panels

- The interface includes five coordinated panels

PCA Scatter Plot

Parallel Coordinates

Summary Statistics

Ranking Bar Chart

Trajectory View

- Users can explore the data through brushing, clicking, filtering, and linked updates

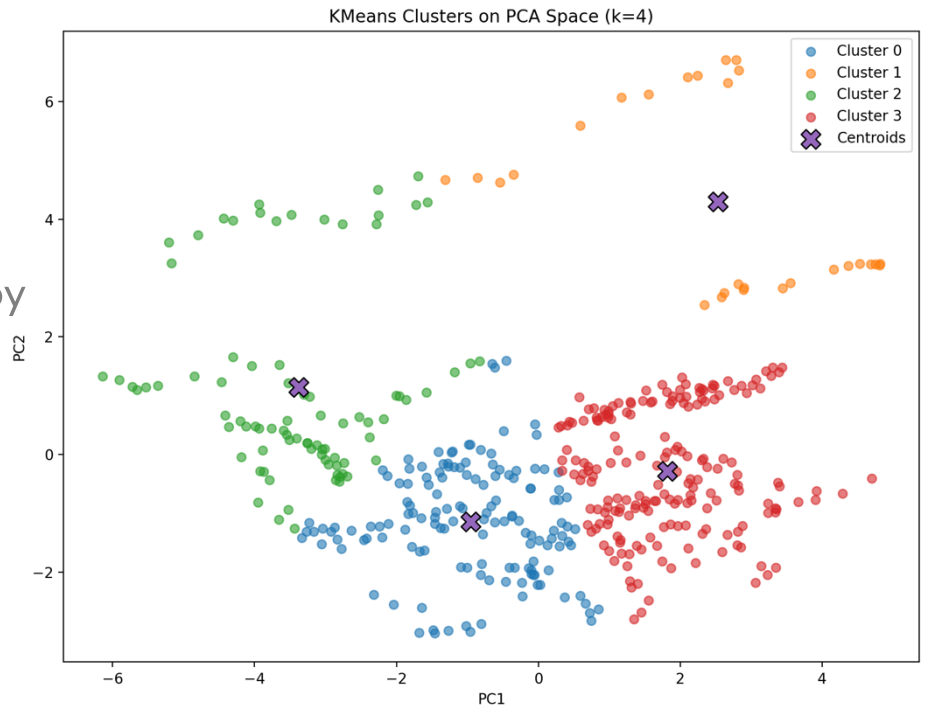
Panel 1 – PCA Scater Plot

Main view of the dashboard

Eachpoint represents one country-year, the color of point can be changed by cluster or region

Axes show PC1 and PC2 :

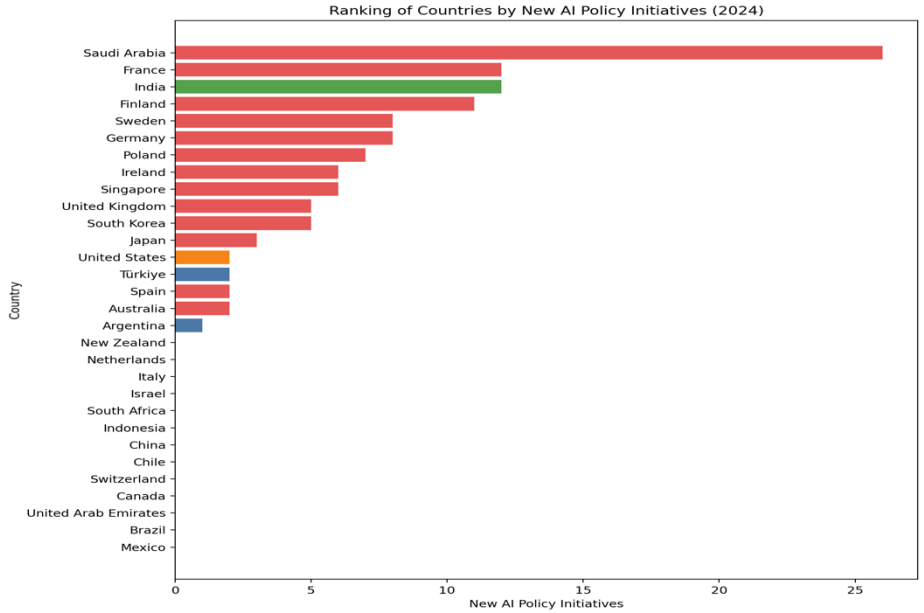
- PC1 mainly captures overall national strength in economic development, research capacity, digital infrastructure, and compute capability.
- PC2 does not represent overall strength, but rather a structural contrast between AI/compute specialization and broader digital-economic maturity.



Panel 2 – Ranking Bar Chart

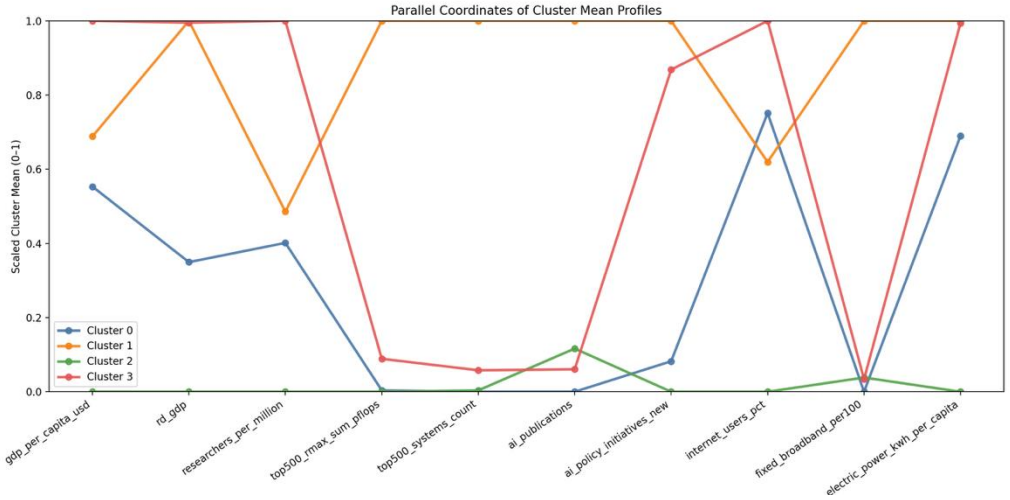
Ranks countries by a selected metric, helps compare countries on specific indicators

Linked with the other 4 panels



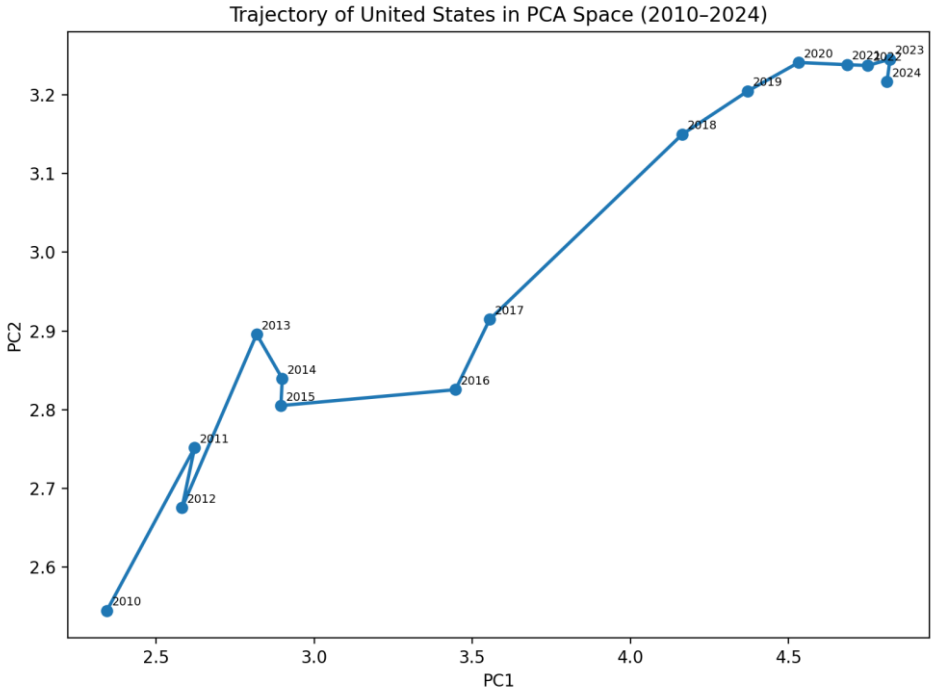
Panel 03 — Parallel Coordinates

Shows the multivariate profiles of clusters
Each line represents one cluster
Helps explain why countries belong to the same group
Supports profile comparison across key variables



Panel 04 — Temporal Trajectory View

how selected countries move in PCA space over time
Each path represents one country from 2010 to 2024
Helps track long-term changes and development patterns
Supports temporal comparison between countries



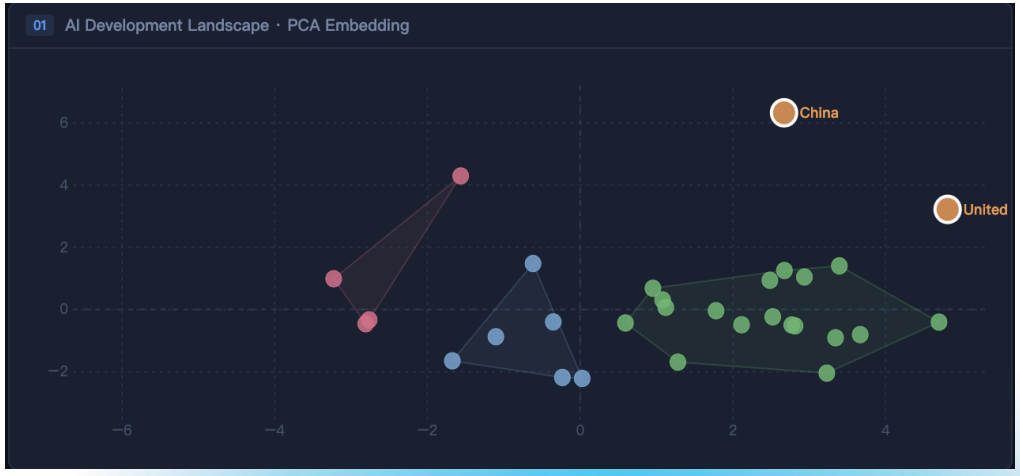
Panel 05 — Selection Summary Statistics

Madantory analytics panel triggered by interaction
Updates automatically after brushing or clicking
Shows summary statistics for the selected subset
Helps compare the selection with the global average

INSIGHTS.

AI Leadership Gap

In 2024, the United States and China form the top AI leadership group.
They are far above the global mean in compute and AI publications.
However, their advantage in AI policy activity is much smaller than their compute advantage.



European Governance Leaders

Several European countries perform strongly in AI policy initiatives.
Their governance activity is high, but their compute capacity is more limited.
This shows a gap between policy leadership and compute leadership.

Europe is positioning itself as the 'Global Regulator' of AI. They lack the massive compute muscle of the US and China, but they own the 'Rulebook' for the digital age.

Emerging Countries Are Improving

India, Indonesia, Mexico, and South Africa gradually move rightward in PCA space.
Their AI publication levels increase over time.
They still remain below the global leaders, but the long-term trend is positive.

GO STRAIGHT to

Insights .

Different Development Paths

China and Germany follow different trajectories in PCA space.
China moves more strongly toward the AI capability / compute direction.
Germany shows a slower path, closer to infrastructure and digital maturity.

Model Linearity: PCA may oversimplify non-linear "leapfrog" developments

Proxy Limitations: Publication counts serve as a proxy for research, but do not fully reflect the "breakthrough quality" or private-sector proprietary innovations.

Part 5

Limitation.

Missing Chip Metrics:

Due to data opacity, semiconductor IP and GPU supply chain data were excluded. This is a key pillar of AI sovereignty not fully captured

Thank you

RONG ZENG · STUDENT ID: 2244780 · ERASMUS